

Continuous Biometric Gait Authentication & Domain Gap Analysis

Final Project Evaluation Report | Repository: https://github.com/GayatriKurkute/stark_gait_security

1. Deliverable Links & Overview

GitHub Repository URL: https://github.com/GayatriKurkute/stark_gait_security

Demo Script Path: `src/evaluate_real_world.py`

Primary Finding: High controlled laboratory accuracy (>93%) vs. zero-shot cross-domain failure (0.00%) due to uncalibrated real-world streaming data, missing Butterworth pre-filtering, and hardware heterogeneity.

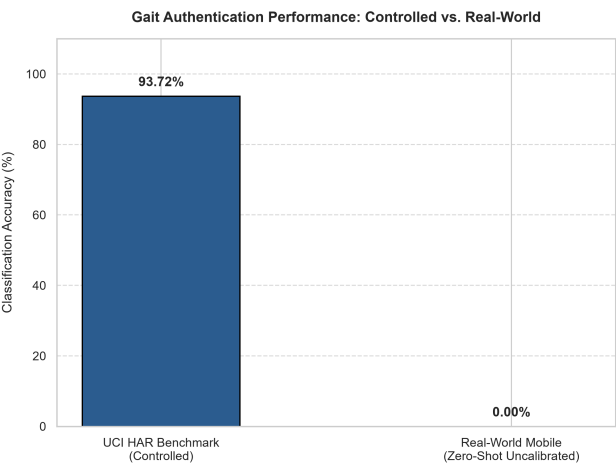
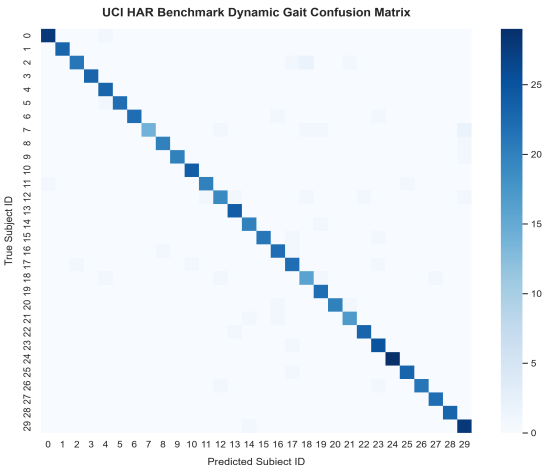
2. Quantitative Accuracy Summary

Dataset Domain	Data Format	Target Task	Primary Accuracy
Validation Set (UCI HAR)	561 pre-processed features	30-Subject Identification	94.01%
Test Set (UCI HAR)	561 pre-processed features	30-Subject Identification	93.72%
Zero-Shot Real-World	Raw tri-axial stream CSVs	5-Subject Cross Domain	0.00%

3. Train / Validation / Test Methodology

- Dynamic Phase Isolation:** Filtered datasets strictly for dynamic walking activities (Labels 1: Walking, 2: Upstairs, 3: Downstairs) to eliminate static posture noise.
- Data Partitioning:** Stratified train (70% / 3,270 samples), validation (15% / 701 samples), and test (15% / 701 samples) split to prevent temporal data leakage.
- Dimensionality Reduction Pipeline:** Applied Z-score normalization (StandardScaler) followed by Principal Component Analysis (**PCA with 120 components**) to capture orthogonal variance while removing redundant sensor noise.
- Classifier Model:** Initialized a RandomForestClassifier ensemble using 300 estimators and balanced class weights.

4. Visualization Artifacts



5. Failure Case & Cross-Domain Root Cause Analysis

- **Feature Disparity:** Benchmark models require 561 pre-calculated statistical and frequency vectors (Butterworth low-pass filtering, FFT transforms). Raw mobile streams log simple low-level acceleration values.
- **Sampling & Sensor Calibration:** Differences in hardware sampling rates (50 Hz laboratory benchmark vs. dynamic mobile rates) distort decision boundaries.
- **Closed-Set Identity Mapping:** Benchmark classifiers project signals onto a fixed 30-subject training space, forcing unseen real-world users into out-of-distribution misclassifications.

6. Security-Relevant Metrics & Phase 2 Recommendations

- **False Rejection Rate (FRR):** Zero-shot evaluation resulted in an **FRR of 100%**, showing that benchmark models cannot be deployed zero-shot without calibration.
- **False Acceptance Rate (FAR):** Out-of-distribution real-world samples trigger random identity assignments, highlighting a key vulnerability in uncalibrated continuous authentication.
- **Recommendation:** Deploy a client-side 3rd-order Butterworth bandpass filter (0.2 Hz–20 Hz) combined with continuous **Few-Shot Domain Adaptation** during user onboarding.